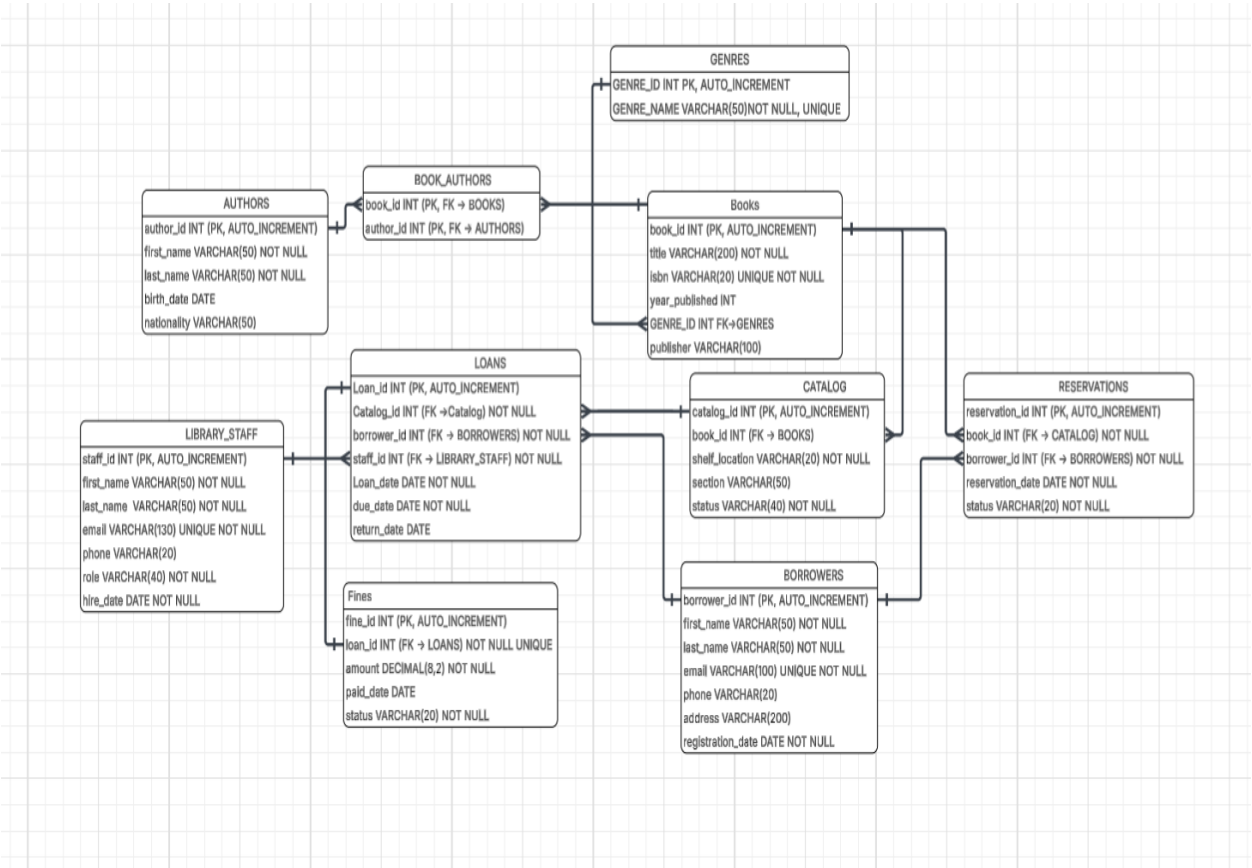
Model Template

1.1 Logical Scheme



1.2 Objects

AUTHORS

Table Name	Field Name	Field Description	Data Type / Constraints
AUTHORS	author_id	Author unique identifier	INT, PK, AUTO_INCREMENT
AUTHORS	first_name	Author first name	VARCHAR(50), NOT NULL
AUTHORS	last_name	Author last name	VARCHAR(50), NOT NULL
AUTHORS	birth_date	Author birth date	DATE
AUTHORS	nationality	Author nationality	VARCHAR(50)

BOOKS

Table Name	Field Name	Field Description	Data Type / Constraints
BOOKS	book_id	Book unique identifier	INT, PK, AUTO_INCREMENT
BOOKS	title	Book title	VARCHAR(200), NOT NULL
BOOKS	isbn	Unique ISBN code	VARCHAR(20), UNIQUE, NOT N
BOOKS	year_published	Year book was published	INT
BOOKS	genre_id	Reference to genre	INT, FK
BOOKS	publisher	Publishing company	VARCHAR(100)

GENRES

Table Name	Field Name	Field Description	Data Type / Constraints
GENRES	genre_id	Genre unique identifier	INT, PK, AUTO_INCREMENT
GENRES	genre_name	Genre name	VARCHAR(50), UNIQUE, NOT N

BOOK_AUTHORS

Table Name	Field Name	Field Description	Data Type / Constraints
BOOK_AUTHORS	book_id	Reference to book	INT, FK
BOOK_AUTHORS	author_id	Reference to author	INT, FK

BORROWERS

Table Name	Field Name	Field Description	Data Type / Constraints
BORROWERS	borrower_id	Borrower unique identifier	INT, PK, AUTO_INCREMENT
BORROWERS	first_name	Borrower first name	VARCHAR(50), NOT NULL
BORROWERS	last_name	Borrower last name	VARCHAR(50), NOT NULL
BORROWERS	email	Borrower email	VARCHAR(100), UNIQUE, NOT I
BORROWERS	phone	Borrower phone	VARCHAR(20)
BORROWERS	address	Borrower address	VARCHAR(200)
BORROWERS	registration_date	Date borrower registered	DATE, NOT NULL

LIBRARY_STAFF

Table Name	Field Name	Field Description	Data Type / Constraints
LIBRARY_STAFF	staff_id	Staff unique identifier	INT, PK, AUTO_INCREMENT
LIBRARY_STAFF	first_name	Staff first name	VARCHAR(50), NOT NULL

LIBRARY_STAFF	last_name	Staff last name	VARCHAR(50), NOT NULL
LIBRARY_STAFF	email	Staff email	VARCHAR(130), UNIQUE, NOT N
LIBRARY_STAFF	phone	Staff phone	VARCHAR(20)
LIBRARY_STAFF	role	Staff role in library	VARCHAR(40), NOT NULL
LIBRARY_STAFF	hire_date	Date hired	DATE, NOT NULL

CATALOG

Table Name	Field Name	Field Description	Data Type / Constraints
CATALOG	catalog_id	Catalog record identifier	INT, PK, AUTO_INCREMENT
CATALOG	book_id	Reference to book	INT, FK
CATALOG	shelf_location	Location of the book on shelf	VARCHAR(20), NOT NULL
CATALOG	section	Library section	VARCHAR(50)
CATALOG	status	Availability status	VARCHAR(40), NOT NULL

LOANS

Table Name	Field Name	Field Description	Data Type / Constraints
LOANS	loan_id	Loan transaction identifier	INT, PK, AUTO_INCREMENT
LOANS	catalog_id	Reference to catalog item	INT, FK, NOT NULL
LOANS	borrower_id	Reference to borrower	INT, FK, NOT NULL
LOANS	staff_id	Reference to staff member	INT, FK, NOT NULL
LOANS	loan_date	Date when book was loaned	DATE, NOT NULL
LOANS	due_date	Return due date	DATE, NOT NULL
LOANS	return_date	Actual return date	DATE

FINES

Table Name	Field Name	Field Description	Data Type / Constraints
FINES	fine_id	Fine identifier	INT, PK, AUTO_INCREMENT
FINES	loan_id	Reference to loan	INT, FK, UNIQUE
FINES	amount	Fine amount	DECIMAL(8,2), NOT NULL
FINES	paid_date	Date fine was paid	DATE
FINES	status	Fine payment status	VARCHAR(20), NOT NULL

RESERVATIONS

Table Name	Field Name	Field Description	Data Type / Constraints
RESERVATIONS	reservation_id	Reservation identifier	INT, PK, AUTO_INCREMENT
RESERVATIONS	book_id	Reference to book	INT, FK, NOT NULL
RESERVATIONS	borrower_id	Reference to borrower	INT, FK, NOT NULL
RESERVATIONS	reservation_date	Date reservation created	DATE, NOT NULL
RESERVATIONS	status	Reservation status	VARCHAR(20), NOT NULL